

CareBridge

The AI prepares. The human decides. Assignment 2: infrastructure, security & responsibility, explainability, metrics & estimation — for the prototype Assignment 1 justified.

Executive summary — the case, and how it should be run

THE PROBLEM

650K prior-auths/yr; ~390K need a licensed UM nurse. Her touch-time goes to **assembling evidence**, not deciding — **\$4.1M**_{/yr} of reviewer labor sits on assembly.

THE INTERVENTION

A RAG-grounded **decision packet**: every criterion met / unmet / uncertain, every assertion cited to source. AI recommends approve / pend / route — **it structurally cannot deny.**

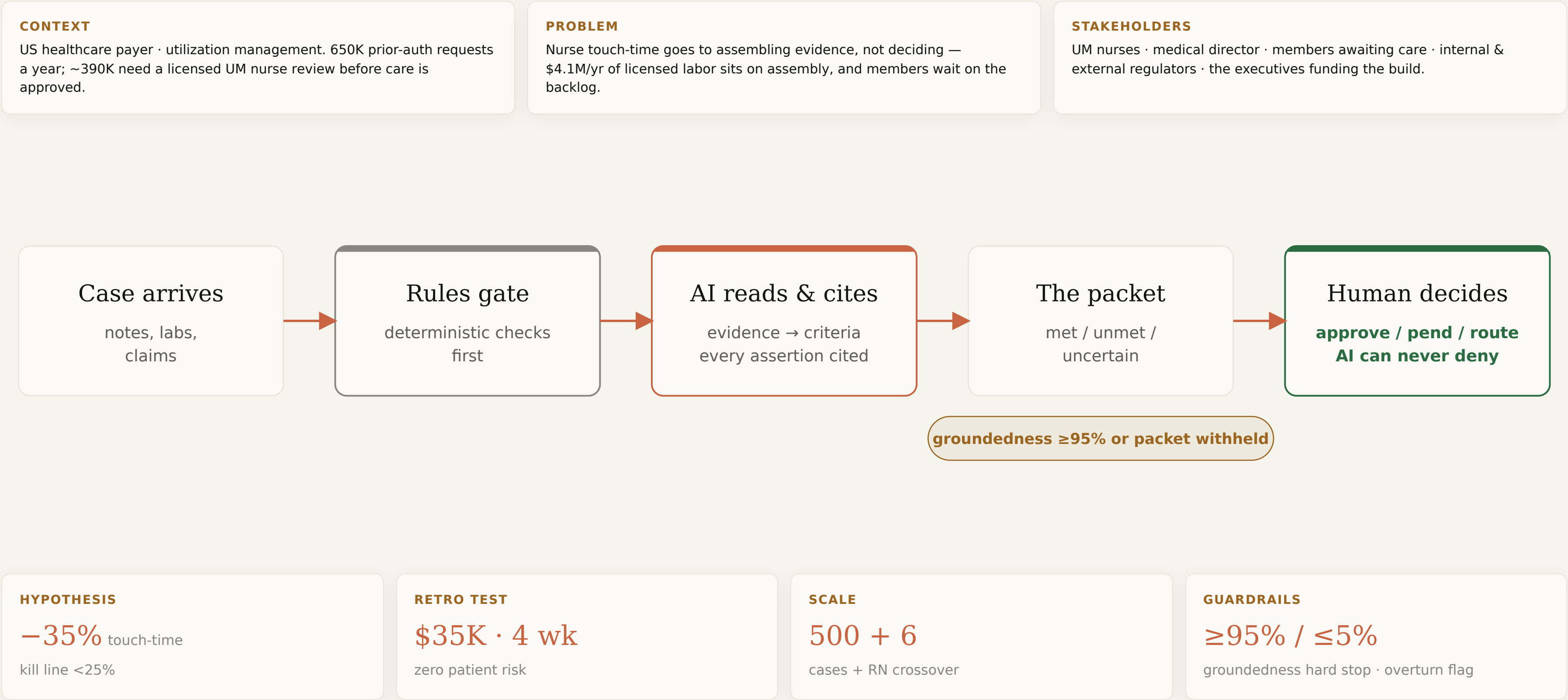
THE ECONOMICS — ANNUAL, AT THE 35% FLOOR (MID COST, B=60%)

Gross labor saving		\$1.43M
Run cost		−\$0.35M
Build (amortized 3 yr)		−\$0.53M
Net annual value		≈\$0.55M

Upside at 45% reduction ≈ \$0.96M. Pessimistic corner ≈ break-even — which is exactly why a \$35K prototype gates the \$1.6M build.

The approach: a \$35K prototype earns MVP scoping only through its gates — infrastructure, security, responsibility and explainability are designed in, not bolted on.

Assignment 1 — the business case, in one picture



The discipline the course grades: **a prototype is not a product** — disposable, fail-fast, validates the business case before the \$1.6M build.

The workflow — rules do the checking, AI does the reading, humans do the deciding



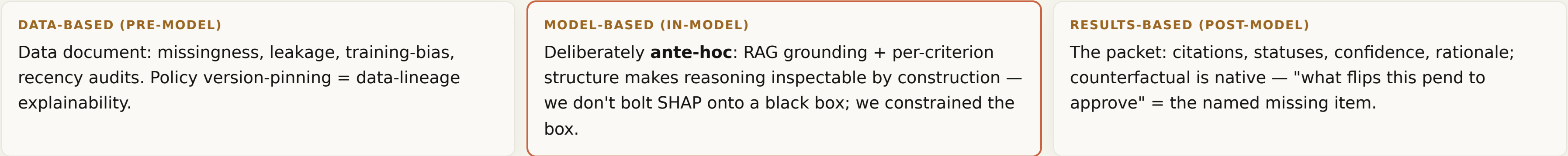
■ Rules / RPA ■ AI ■ Human authority

Primary: UM nurse Secondary: medical director / exceptions Tertiary: UM ops executive (read-only)

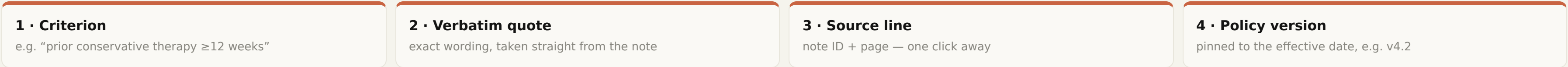
Explainability is stakeholder-specific — one system, five different "why"s

STAKEHOLDER	THEIR QUESTION	THE EXPLANATION THEY GET
UM nurse	Why does it say criterion 2 is met?	Cited source span (note ID + page), met/unmet/uncertain status — verifiable in seconds
Medical director	Why was this routed to me?	The unresolved criterion + the evidence gap, not a score
Compliance / auditor	Can you defend this decision from record?	Immutable audit trail: every assertion → source → human action
Member / provider	Why was I pended?	The single named missing item (e.g., "TB screening result") — actionable, not opaque
Executive	Is the system behaving?	Groundedness %, overturn trend, touch-time — population-level explanation

Where CareBridge sits in the explainability taxonomy

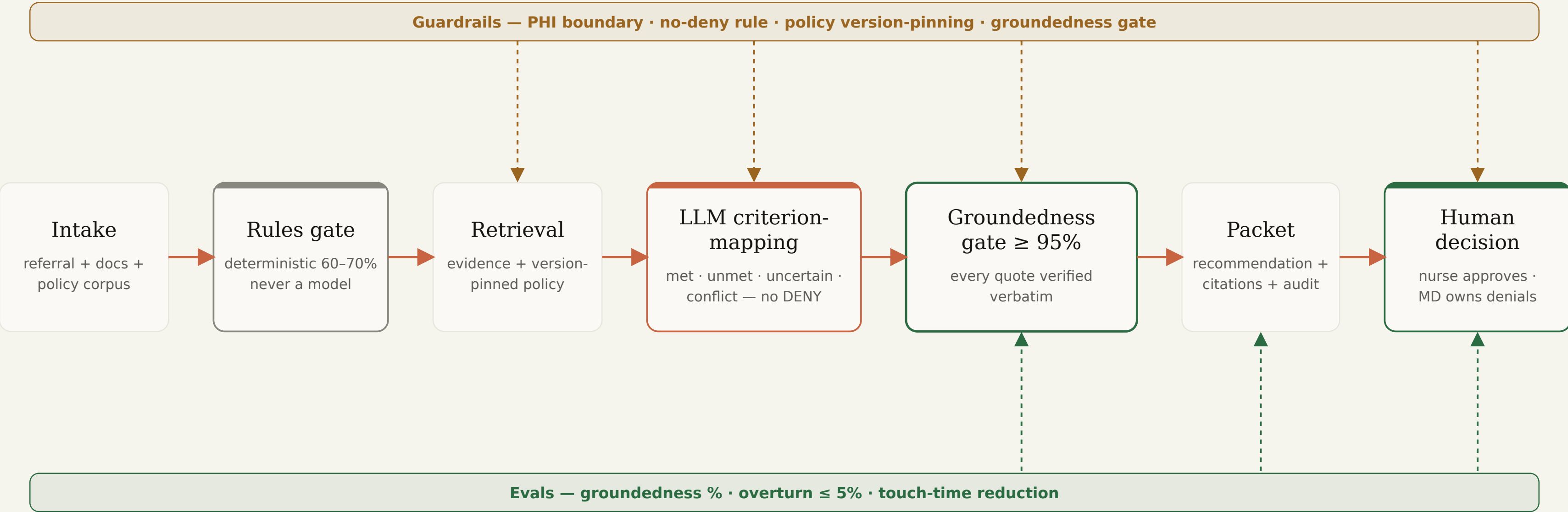


Corner cases (the brief asks): conflicting notes → criterion surfaces BOTH spans, statused *conflict*, routed to MD; missing policy version → hard fail to human, never silent fallback; low retrieval confidence → packet declares evidence-completeness "medium/low" instead of bluffing. All three visible in the demo.



This chain is why the packet needs no separate “explain” button — every status is traceable end to end, criterion to source. Full walkthrough: appendix.

System design — one pipeline, three kinds of worker



CPU or GPU? The latency budget decides — and it's hours, not milliseconds

THE LATENCY BUDGET, TO SCALE



Rules / RPA

the deterministic 60–70%: eligibility, codes, benefit — never a model

CPU + managed API

de-identified criterion reasoning; prototype is CPU-only overnight batch

GPU — one place only

on-prem PHI-side embedding/retrieval, because PHI can't leave

Paying for millisecond GPU serving against a multi-day SLA is buying speed nobody asked for. Delayed/queue-ahead processing is the honest answer — packet ready before the nurse opens the case.

Fermi sizing — done aloud

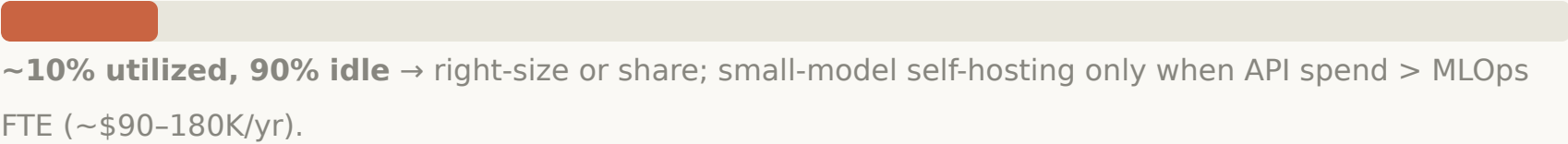
STEP	ARITHMETIC	RESULT
Daily volume	390K/yr ÷ 260 days	~1,500 cases/day
Tokens/case	~30 pages ≈ 40K in + 2K out	~42K tokens
Daily tokens	1,500 × 42K	~63M/day
API cost	63M/day × blended rate	~\$60–75K/yr ✓ matches A1 band
Embedding GPU	63M ÷ ~10K tok/s	~1.75 GPU-h/day
Dev time	bottom-up	MVP 2–3 quarters, \$1.2–2.0M

AI COST VS THE LABOR IT COMPRESSES (PER CASE)

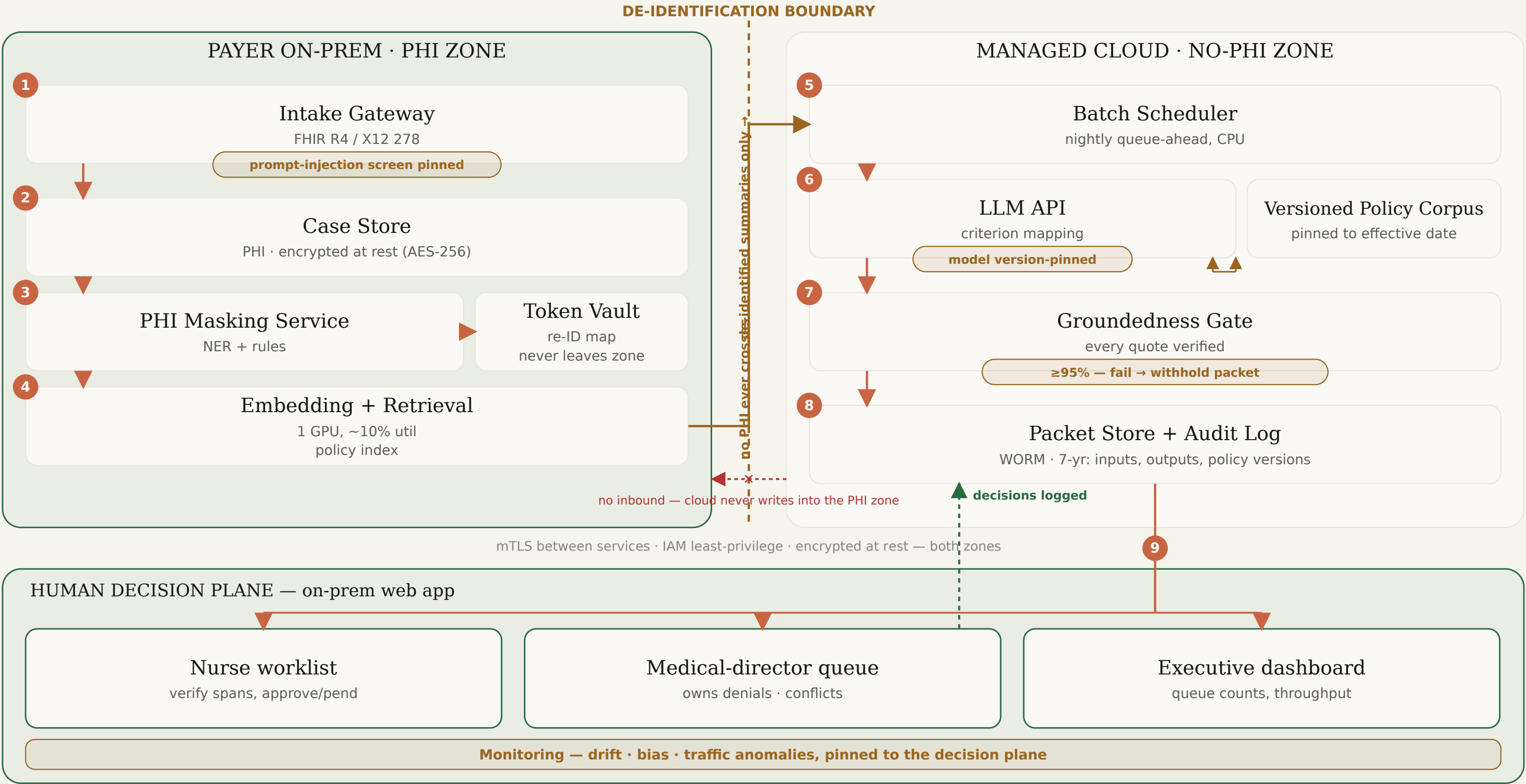


≈2% overhead on the cost it removes 35% of.

ONE GPU'S DAY

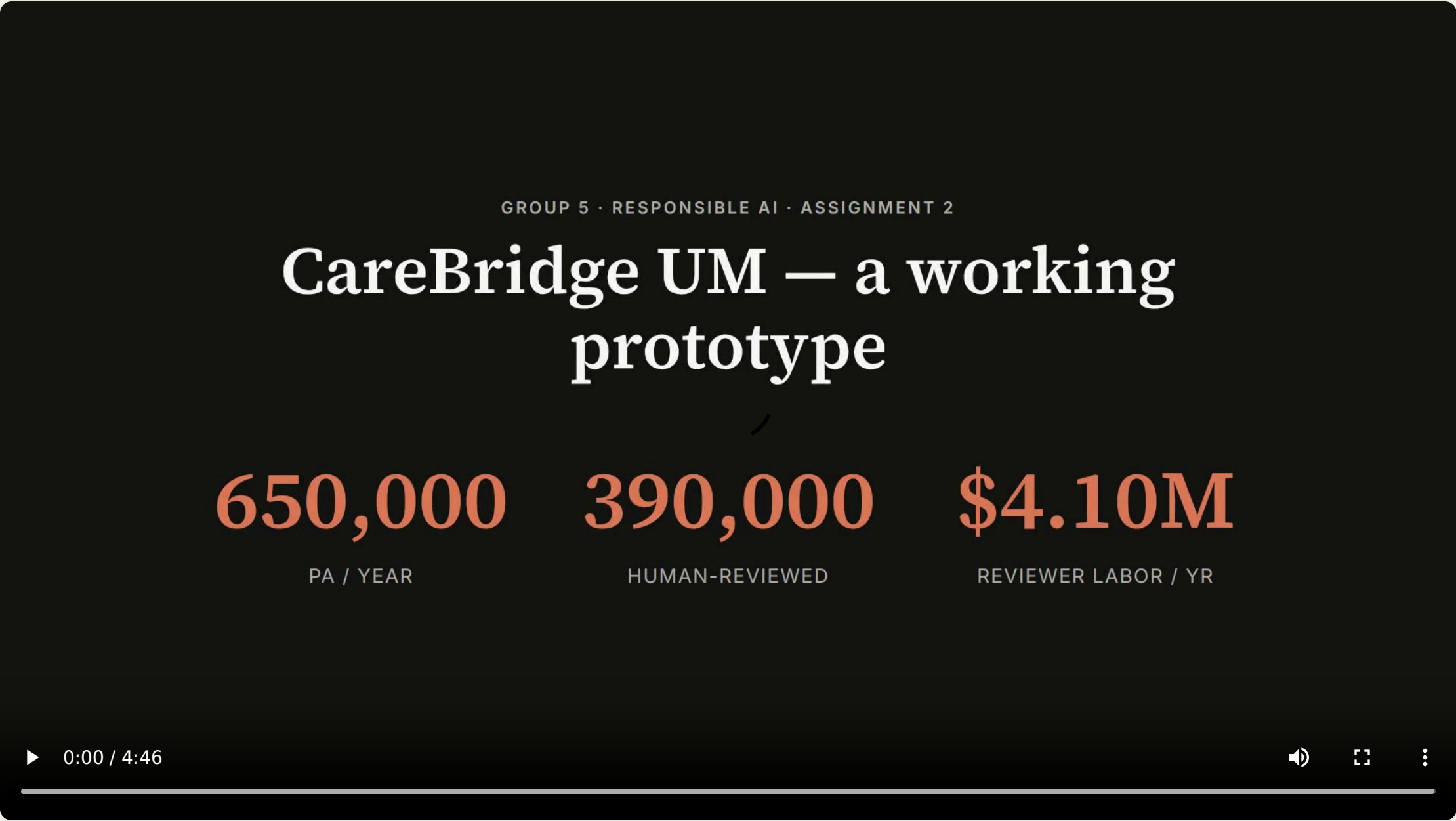


Same two zones, at engineering resolution



Every hop labeled; PHI never crosses the amber line — the packet is the only thing that travels.

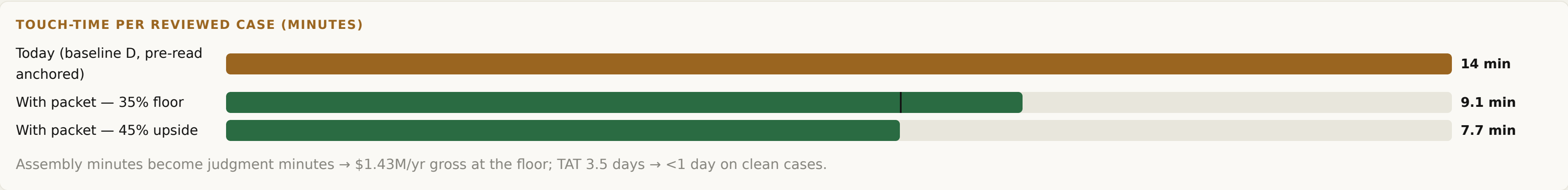
Demo — CareBridge, end to end



4 minutes 46 · narrated by all three of us

Video won't play? Fallback: [video.html](#)

Productivity gain without replacement — the engagement design

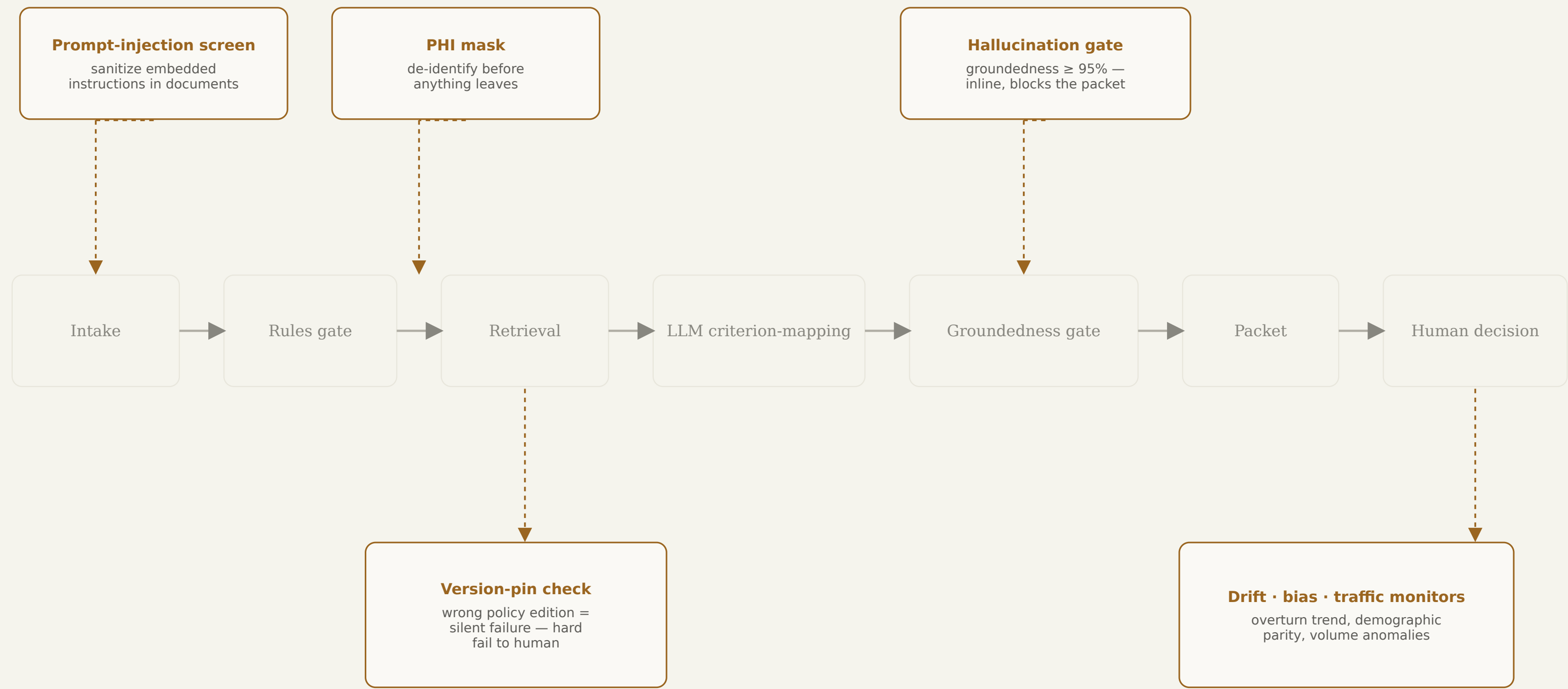


- **Enabler, not replacer — by structure, not slogan:** the AI cannot deny; its ceiling is pend/route. Human authority is a design constraint, not a policy promise.
- **Engagement mechanics:** reviewers co-design the packet layout · disagreement button on every packet (override = training signal, not insubordination) · savings published back as capacity, not quota · judgment-ladder career path (complex cases, audits, appeals).

Responsibility, per stakeholder

STAKEHOLDER	RESPONSIBILITY QUESTION	CAREBRIDGE ANSWER
Developers	Importing bias from historical denials?	Historical labels = efficiency benchmark only, never fairness; stratified reporting; flags → powered fairness study
End users (nurses)	Can I verify what it claims?	Cited spans + statuses; groundedness ≥95% hard stop
Medical director	Do I still own denials?	Structurally yes — AI cannot deny
Internal regulators	What's masked, what's tracked?	PHI masked on-prem before any LLM call; immutable audit — inputs, outputs, policy versions, 7 yr
External regulators	Defensible record per decision?	Packet + audit trail per case
Sponsors	Real savings, engaged team?	Powered endpoint; engagement mechanics; redeployment plan
Members	Safer or worse off?	Overturn ≤5% guardrail; no automated denial, ever

Security — every monitor pinned where the threat enters

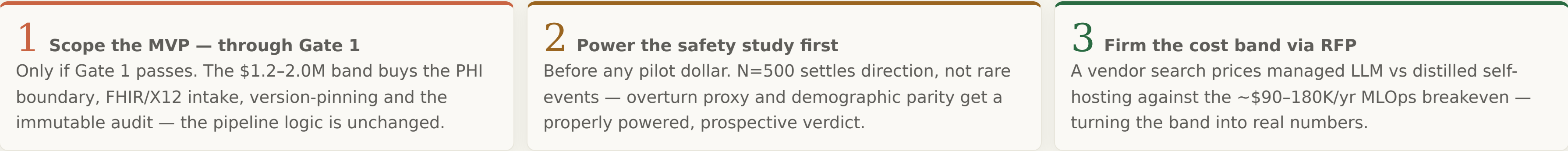


The path forward — every stage earns the next



Every gate is a stop condition committed in advance — **four cheap ways out (two inside Gate 1), not one expensive way in.**

THREE NEXT STEPS THIS APPROACH PRESCRIBES — EACH CONDITIONAL, NONE AUTOMATIC



STILL A PROTOTYPE — synthetic data only · no PHI · disposable build · fidelity lives in the explanation, not the engineering · \$35K gate before \$1.6M

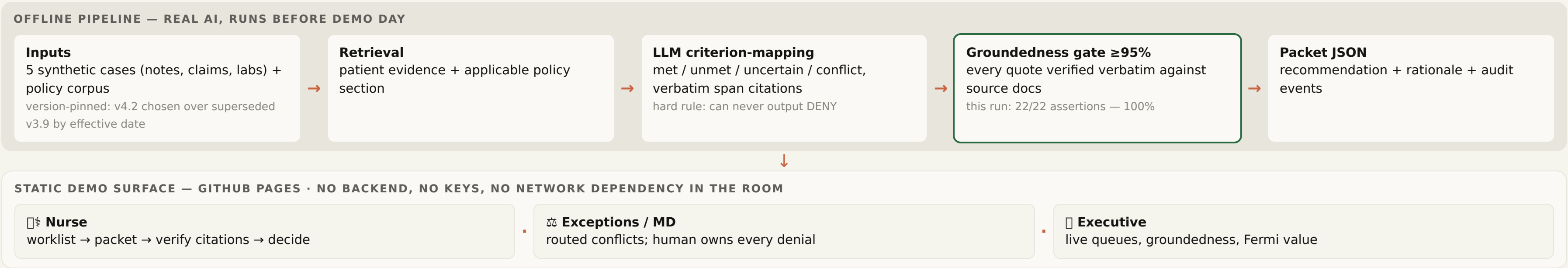
The packet is the explanation. The human owns the decision. The prototype gates the money.

Group 5 · Q&A — explainability & demo: Sanjay · cost & scale: Thejas · bias & stakeholders: Ashish

Archive — backup slides

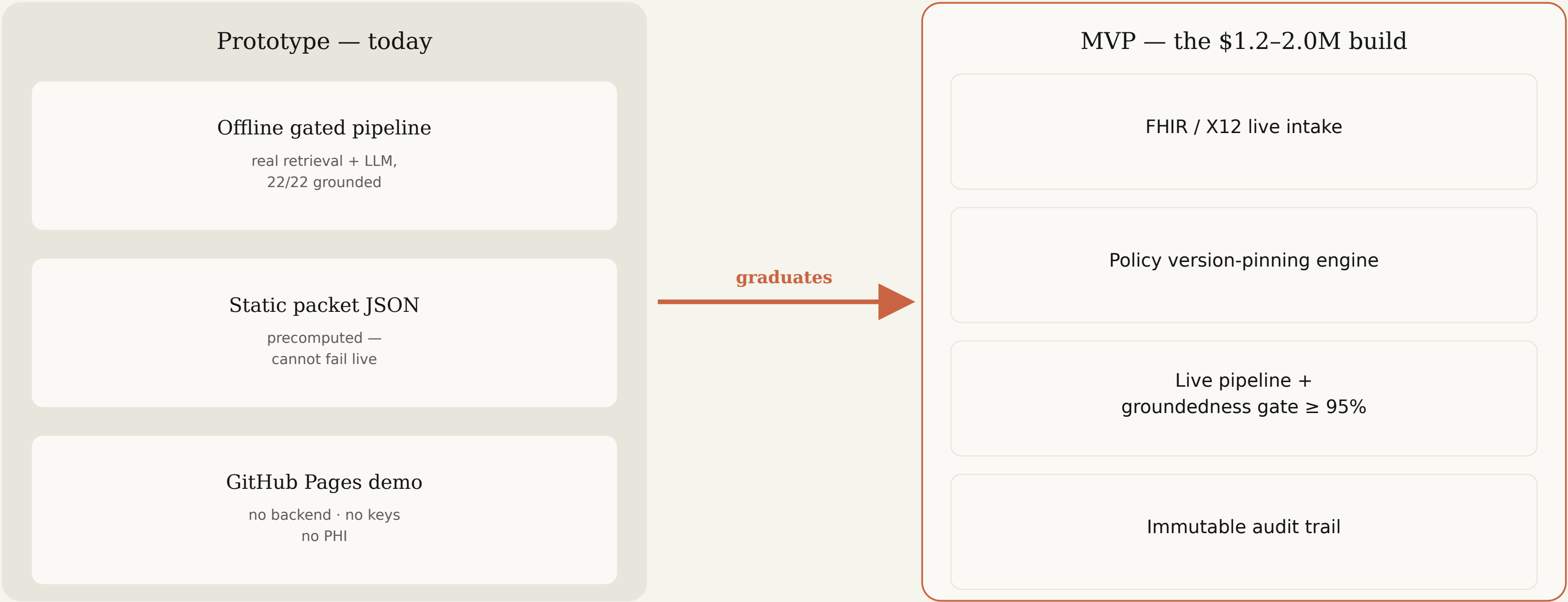
Detail we can pull up in Q&A.

Architecture — precomputed-real: the pipeline is genuine, the demo cannot fail



MVP boundary (the A2 build): PHI-side retrieval/embedding on-prem/VPC; only de-identified criterion summaries reach the managed LLM; FHIR/X12 integration, policy version-pinning engine and immutable audit are what the \$1.2–2.0M estimate buys.

Prototype → MVP: same pipeline, graduated surroundings



The pipeline logic and gates are identical — the MVP buys integration, live-ness, and auditability. 2-3 quarters.

Data — what feeds it, how clean, where it lives

DATA	EXAMPLES	CLEANLINESS
Request envelope	member/plan ID, NPI, CPT/HCPCS, ICD-10, urgency	4.5 / 5 — structured
Clinical evidence	progress notes, discharge summaries, labs, rx history	2.5 / 5 — unstructured, OCR-contaminated
Policy corpus	plan medical policies (version-pinned), CMS NCD/LCD	3 / 5 — version drift = silent failure
Outcome labels	historical adjudication, appeals, overturns	4 / 5 — test ground truth

- Overall 3/5 — set by exactly the two inputs the AI must handle (notes + policy versions).
- PHI on-prem only, HIPAA limited data set under DUA. Public proxies for build: MIMIC-IV, CMS MCD, Synthea. InterQual/MCG excluded — no license dependency.

360° — how CareBridge answers every A2 topic

TOPIC	THE ONE-LINE ANSWER	DEPTH
Explainability	The packet <i>is</i> the explanation — every assertion cited, statused, auditable per stakeholder.	Deep-dive
Infrastructure	CPU/batch prototype; hybrid MVP (on-prem GPU for PHI retrieval, managed LLM for de-identified reasoning); latency budget is hours, not ms.	Deep-dive
Responsible AI	Capacity release not replacement; per-stakeholder responsibility grid; AI structurally cannot deny.	Deep-dive
Security	Monitor hallucination (groundedness), drift, prompt-injection, bias spikes, traffic anomalies.	Covered · slide 20
Performance metrics	Touch-time (\$ via Fermi), groundedness $\geq 95\%$, approve-precision $\geq 95\%$, overturn $\leq 5\%$ — each linked to a business goal.	Covered · slide 20
Estimation	Fermi: 390K cases/yr $\rightarrow$ 63M tokens/day $\rightarrow$ ~\$60–75K/yr API + ~1 mostly-idle GPU; build \$1.2–2.0M, 2–3 quarters.	Covered · slide 14

Demo: the packet is the explanation

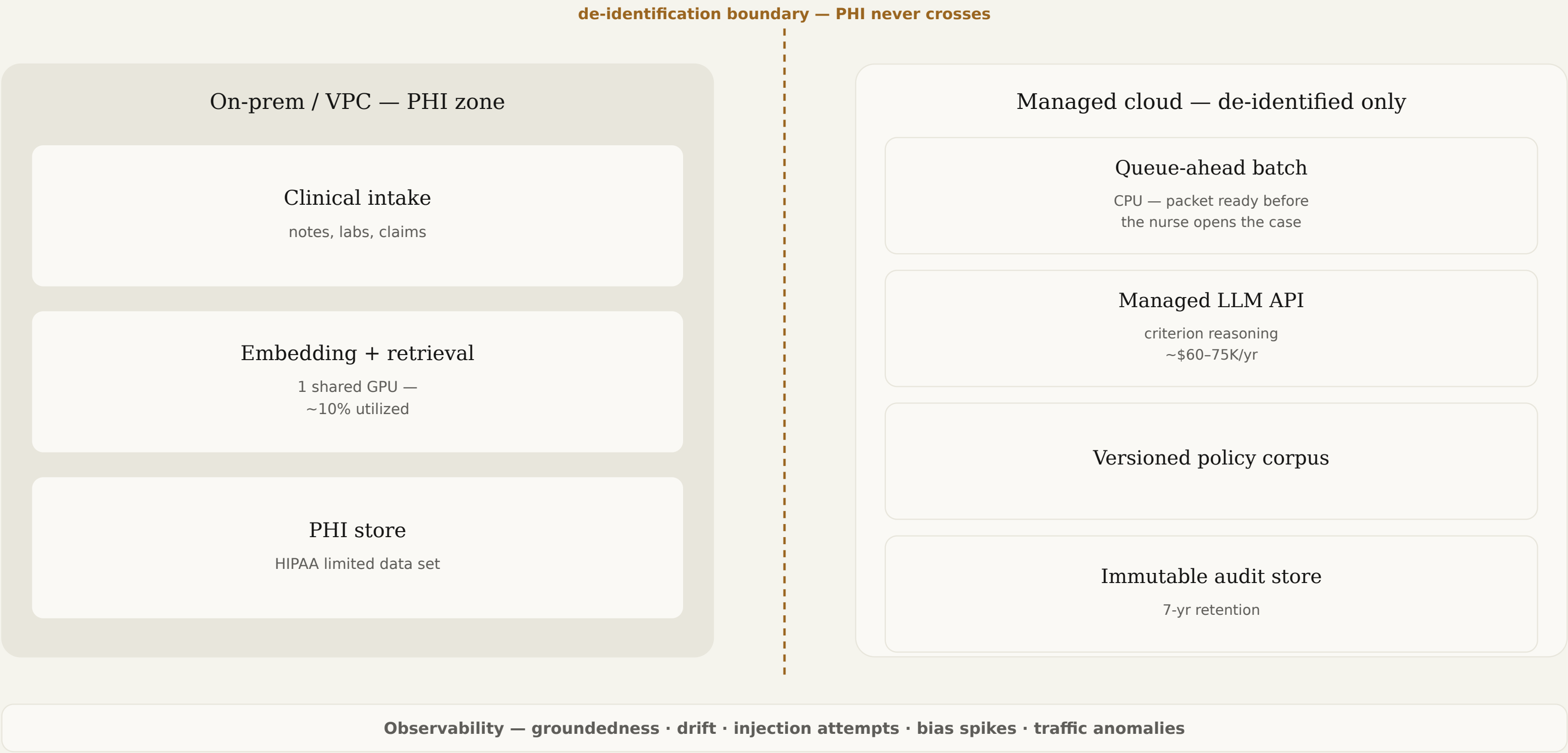
Nurse flow
worklist → adalimumab case → 2 criteria met (cited), 1 uncertain (TB screen missing)
→ PEND with the named gap

Exceptions flow
conflict case routed → medical director decides → the denial button exists only here

Executive dashboard
live queues, groundedness, capacity — population-level explainability

Every packet in this demo was generated by a real retrieval + LLM pipeline before this session — 22/22 assertions verified verbatim against source documents. Nothing on screen is hand-written.

Deployment — a PHI zone, a managed zone, and one honest boundary



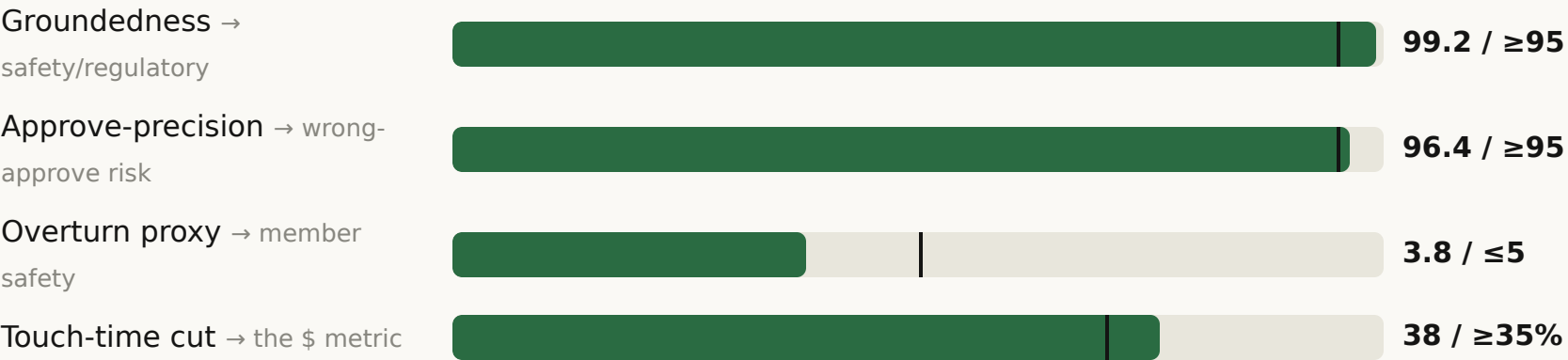
Latency budget is hours, not milliseconds — queue-ahead beats real-time GPU serving nobody asked for.

Security monitoring & metrics — linked to business goals

MONITOR (PER BRIEF)

Hallucinations	groundedness per assertion, $\geq 95\%$ veto
Response drift	weekly fixed canary set, diffed
Prompt injection	notes = untrusted input → strict templates, no tool authority, injection evals
Bias/toxicity	stratified concordance dashboards
Latency/volume	queue anomaly alerts

METRIC VS TARGET — AND THE BUSINESS GOAL IT PROTECTS



Ticks = targets. Overturn is directional at N=500 — powered follow-up before pilot.